Implementation of 2D-DCT Based on FPGA with Verilog HDL

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Abstract. Discrete Cosine Transform is widely used in image compression. This paper describes the FPGA implementation of a two dimensional (8×8) point Discrete Cosine Transform (8×8 point 2D-DCT) processor with Verilog HDL for application of image processing. The row-column decomposition algorithm and pipelining are used to produce the high quality circuit design with the max clock frequency of 318MHz when implemented in a Xinlinx VIRTEX-Ⅱ PRO FPGA chip.

Keywords: 2D-DCT, FPGA, Verilog HDL.

1 Introduction

Discrete Cosine Transform (DCT), among various transforms, is the most popular and effective one in image and video compression. So far, it has been adopted by many international standards like JPEG, MPEG1, MPEG2, MPEG4, and H26x, etc. However, it is difficult to satisfy the requirement for real-time by software owing to its heavy quantity of computing. Therefore, a hardware method is adopted to satisfy the requirement for speed in many practical applications. Hardware implementation algorithms of 2D-DCT can be divided into two categories: row-column decomposition methods and non row-column decomposition methods [1-2]. 4096 times of multiplications and 4032 times of additions are required for the 8×8 point 2D-DCT calculation. The row-column decomposition method is usually adopted to reduce the computational complexity. In this paper, the row-column decomposition algorithm which changes 2D-DCT to two 1D-DCT (one dimensional point Discrete Cosine Transform) is used. In the design of 1D-DCT, direct multiplication operation has been replaced by pre-adder, partial multiplication based on look-up table, partial multiplication result shift and addition so as to reduce the hardware cost.

2 DCT Algorithms

In this section the 1D-DCT and 2D-DCT will be presented. For the 1×8 sequence $f(i)$, DCT can be defined as:

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$$F(k) = \frac{1}{2} C(k)[\sum_{i=0}^{7} f(i) \cos \frac{(2i+1)k\pi}{16}] \tag{1}$$

Where k = 0,1,2……7,C(0)=1/$\sqrt{2}$ (t=0), C(t)=1 (t≠0).

The equation for 2D-DCT of a (8×8) input sequence *f(x,y)* representing the image pixel values is shown below [3].

$$F(\mathrm{u,v}) = \frac{1}{4} C(u)C(v)[\sum_{x=0}^{7}\sum_{y=0}^{7} f(x,y)\cos\frac{(2x+1)u\pi}{16}\cos\frac{(2y+1)v\pi}{16}] \tag{2}$$

Where u,v = 0,1,2……7; C(0)=1/$\sqrt{2}$ (t=0), C(t)=1 (t≠0).

3 Hardware Implementation of 2D-DCT

This section presents the implementation of the DCT on FPGA in detail.

3.1 Design Strategy

According to the definition of 2D-DCT above, the equation for 2D-DCT can be represented as following.

$$F(\mathrm{u,v}) = \frac{1}{2} C(u)\sum_{x=0}^{7}[\frac{1}{2} C(v)\sum_{y=0}^{7} f(x,y)\cos\frac{(2y+1)v\pi}{16}]\cos\frac{(2x+1)u\pi}{16} \tag{3}$$

It can be introduced from (3) that 2D-DCT can be decomposed into two 1D-DCT, so the fast algorithm for computing 1D-DCT can be used for 2D-DCT. The matrix $F=CfC^T$ can be used to rewrite (3), where C is on behalf of the 8×8 2D-DCT coefficient matrix [4]. It can be deduced from the matrix expression that the 2D-DCT calculation results can be obtained by the following four steps:

(1) calculating Y=Cf .
(2) transposing the matrix Y to get Y'.
(3) calculating F= CY'.
(4) transposing the matrix F to get the final result.

With row-column decomposition, the 8 point 1D-DCT is applied to each row of the input 8×8 pixel matrix, and a further 1D-DCT is applied to each column of 8×8 block of semi-transformed values. The final results of 2D-DCT can be achieved by transposing the 8×8 matrix which is obtained by 8 point 1D-DCT twice. The implementation architecture of 2D-DCT is shown in Fig.1. The 2D-DCT module includes a control module, data selection module, serial to parallel module, 1D-DCT module, parallel to serial module, memory matrix RAM module, bit interception module and transposing address generation module. In this paper, the 1D-DCT module is reused to meet the speed requirement and reduce hardware resource cost.